

Assignment: Analyzing Sociotechnical Failures in Healthcare

Assignment Overview

This assignment requires you to apply the principles of sociotechnical systems to a practical scenario. You will examine how software functions as an essential component of a broader system that serves a human, social, or organizational purpose. Using concepts such as emergent properties, non-determinism, and system defenses, you will evaluate why a system may be perceived as a success by one group of stakeholders while failing another.

Assignment Description

A large hospital recently implemented MediSync, a new electronic health record system.

- **The Goal:** The organization aimed to improve data management, comply with new government healthcare regulations at the society layer, and increase revenue through accurate billing at the organization layer.
- **The Conflict:** To satisfy reporting goals, the application requires nurses to complete numerous mandatory digital forms for every patient interaction, a process involving people and computer systems to support business activities.
- **The Outcome:** Management considers the system a success because financial reports are accurate and the middleware provides perfect access to remote databases. However, nurses report spending significantly less time on direct patient care, leading to a reduction in the quality of care. Last week, a critical medication error occurred because a nurse, pressured by data-entry requirements, overrode a system warning.

Your task is to write a System Analysis Report evaluating the MediSync system using the sociotechnical systems stack and systems engineering principles.

Assignment Instructions

1. Identify specific elements from the MediSync scenario for at least four layers of the stack, such as equipment, application systems, business processes, or the organization.(7mks)
2. Discuss how properties like reliability or usability have changed since the system was integrated. Explain why these properties can only be assessed and measured once the components are integrated into the whole system.(10mks)
3. Contrast the managerial perspective of success with the clinical perspective.
 - Explain why this scenario represents a wicked problem where a complete specification is not possible and success is a judgment rather than an objective measurement.(10mks)
4. Human Error and System Defenses:
 - Explain the medication error using the systems approach, which assumes people are fallible, rather than the person approach which places blame on the individual.(6mks)

- Using the Swiss cheese model of system failure, suggest two system defenses or barriers that could be implemented to detect errors before they lead to failure.(5mks)
- 5. Discuss how decisions made during the procurement stage, such as purchasing off-the-shelf components, might have limited design options or influenced the system's dependability. (12mks)

Assignment Tips

- Remember that interactions and dependencies exist between layers, and changes in regulations at the society level ripple through the business processes and application software.
- Complexity arises because the properties and behavior of system components are inextricably inter-mingled.
- Systems that include humans are non-deterministic and do not always produce the same output because human behavior is not always consistent.